

Project — Fit a Leaky Neuron

Time to finish: 1 day (about 4-6 hours, working alone or in a pair).

What you will do

You are given voltage recordings of one neuron. Your job is to fit a simple Leaky Integrate-and-Fire (LIF) model to it and check that the model behaves like the real neuron.

Background

The LIF model treats a neuron as a leaky capacitor:

$$\tau \cdot dV/dt = -(V - V_{\text{rest}}) + R \cdot I$$

- τ (τ) is the membrane time constant (how fast V decays).
- R is the membrane resistance (how strongly the neuron reacts to input).
- When V hits a threshold, the neuron spikes and V resets.

You will estimate τ and R from the data.

Data

Everything you need is in `shared/data/`:

- `voltage_traces.npy` — voltage over time at several input currents.
- `spike_times.csv` — when the neuron spiked in each trial.
- `stimulus_protocol.csv` — which current was injected in each trial.

Steps

1. Load and look. Open the notebook `student/notebooks/00_starter.ipynb`. Load one voltage trace and plot it. Understand what one trial looks like.
2. Fit τ and R from subthreshold data. Pick trials where the neuron did NOT spike. Fit $V(t)$ with the closed-form LIF solution $V(t) = V_{\text{rest}} + R \cdot I \cdot (1 - \exp(-t/\tau))$ using `scipy.optimize.curve_fit`. Report your τ (in ms) and R (in $M\Omega$).
3. Predict the firing rate. Using your fitted τ and R , predict the firing rate (spikes per second) for each input current. Compare with the measured rates.
4. One-figure result. Make one plot: measured firing rate (dots) vs. predicted firing rate (line), across all currents. Write 3-4 sentences saying how well the model fits and where it fails.

What to hand in

- Your filled notebook.
- The one plot from Step 4.
- Your fitted τ and R values.
- A short paragraph (3-4 sentences) on how well the model works.

If you finish early, try Step 2 on a different subset of trials and see if your τ and R change — that already tells you something about uncertainty.